

UCE406 :DESIGN OF CONCRETE STRUCTURES

L	T	P	Cr
3	1	2*	4.0

Course Objective: To expose the students to design philosophies & methodologies of various methods of design for reinforced concrete elements

Introduction: Reinforced concrete, definition, properties of materials, grades of concrete and reinforcing steel, stress-strain curves, permissible stresses, design philosophies: working stress design, ultimate strength and limit state design method.

Limit State Design Method: Introduction, Limit States, Characteristic values, characteristic strength, characteristic loads, design values for materials and loads, factored loads.

Limit State of Collapse (Flexure): Types of failures, assumptions for analysis and design of singly reinforced, doubly reinforced, and flanged sections, Design of Lintels, Design of one-way slabs and two-way rectangular slabs.

Limit State of Collapse (Shear, bond and torsion): Introduction - Design for shear, structural components subjected to torsion, design of rectangular beam section for torsion, development length, the continuation of reinforcement (beyond cut-off points).

Limit State of Serviceability: Deflection, effective span to effective depth ratio, modification factors for singly reinforced, doubly reinforced and flanged beams, crack formation and its control

Limit State of Collapse (Compression): Columns and their classification, reinforcement in columns, assumptions, short and long (both tied and helical) columns subjected to axial load, short columns subject to axial, uniaxial and biaxial bending, Interaction Diagrams

Limit State Design of miscellaneous structures: Design of isolated footings, Design of staircases.

Introduction to Working Stress Design Method

Application of SP 16 and Detailing of Reinforcement: Use of SP: 34, Codal Provision for RC Elements: (I) General (II) for ductility.

Laboratory Work (if applicable)

Project would be based on “Design of Concrete Mixes of Different Grades, Study of Strength Properties and Flexural Behaviour of RCC Beams”

1. Design and development of Concrete Mix of a particular Grade of concrete
2. At the age of 28 days, measurement of properties such as Compressive Strength, Splitting Tensile Strength, Flexural strength, and Modulus of Elasticity. Plot the stress-strain curve of concrete at 28 days of age and Reinforcing Steel
3. Study of behaviour of any one type of RCC beams made of the same grade of concrete, subjected to flexural loading.
 - a. Balanced Section
 - b. Under-reinforced Section
 - c. Over-reinforced Section

Course Learning Objectives (CLO)

Upon completing this course, the students can:

1. Analyse the flexural members.
2. Design and Detail the flexural member with consideration for deflection, shear, bond and torsion
3. Design and detail compression members
4. Design other elements such as footings, stair-case

Text Books

1. Reinforced Concrete Design, Gambhir, M. L., Prentice Hall of India(2013).
2. Limit State Design of Reinforced Concrete, Jain, A. K., Arun K. Jain, Dr. Punmia, Laxmi Publications (2007).
3. Limit State Design, Ram Chandra, Standard Book House (2011).
4. Limit State Design of Reinforced Concrete Structures, P. Dayaratnam, P Sarah, Meditech, 2nd Edition, 2017
5. Design of Reinforced Concrete Structures, N. Subramanian, Oxford University Press, (2013)

Reference Books

1. Reinforced Concrete Design, Pillai & Menon, Tata McGraw Hill Publishers(2022).
2. Limit State Design of Reinforced Concrete, Varghese, P. C., Prentice Hall of India(2008).
3. Fundamentals of Reinforced Concrete, Sinha, S. N. and Roy, S Chand Publishers(2014)

Evaluation Scheme

Sr. No.	Evaluation elements	Weightage (%)
1	MST	25
2	EST	35
3	Sessional: (May include the following) Assignment, Regular Lab assessment and Quizzes, Project (Including report, presentation etc.)	40